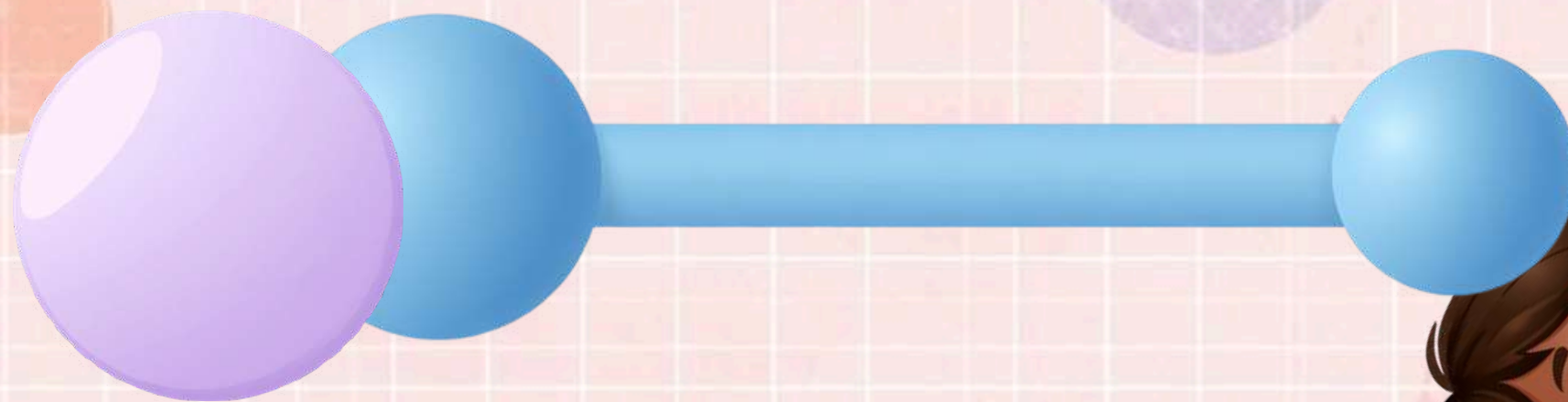
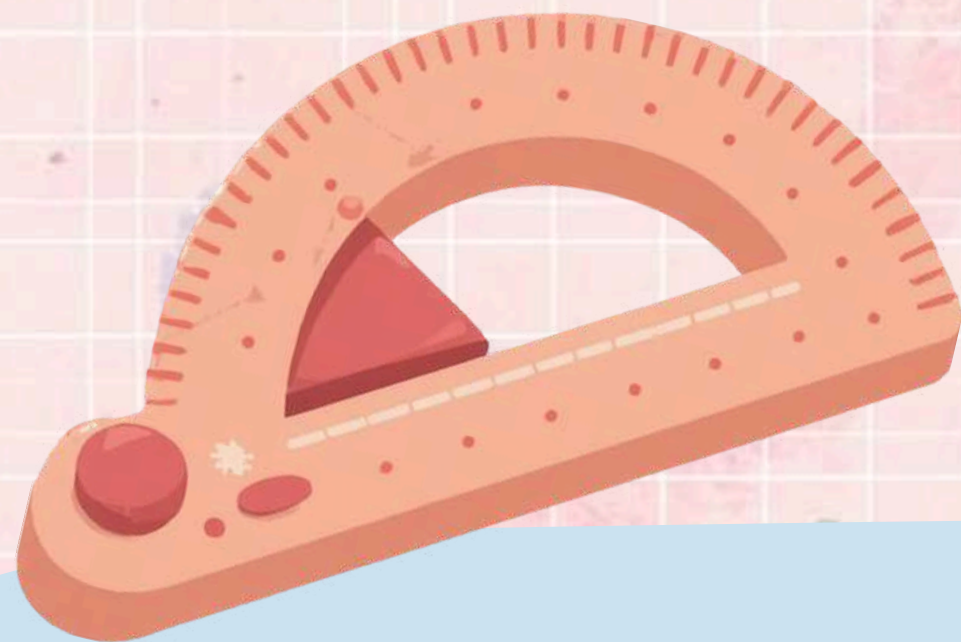
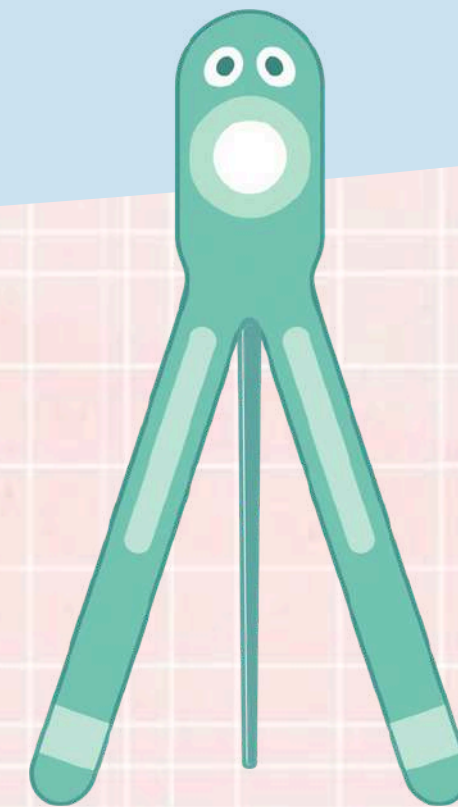
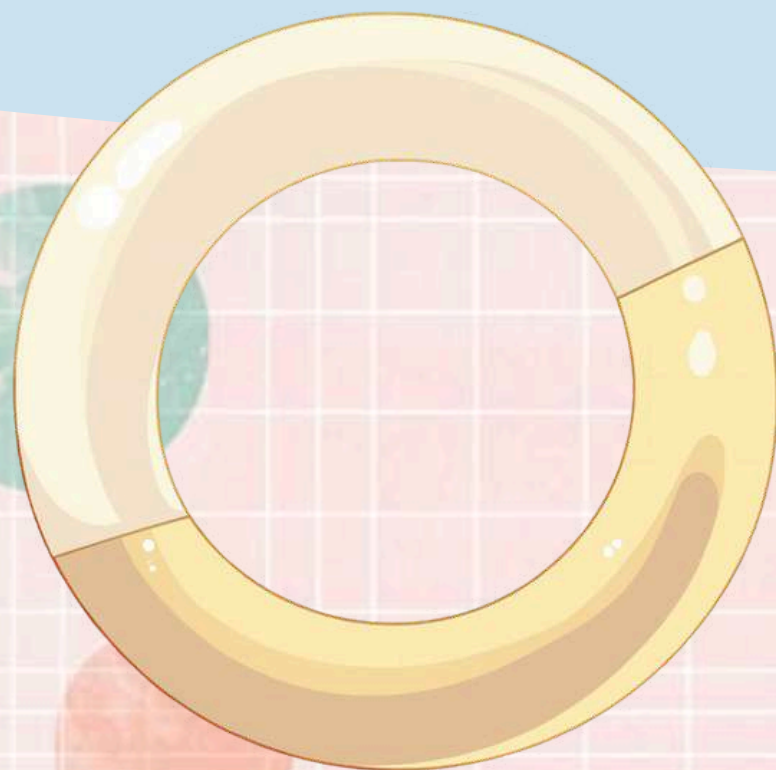




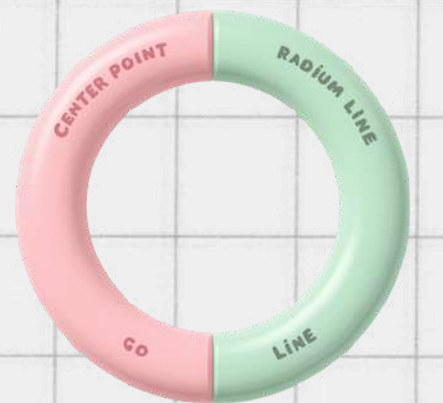
Circle Theorems: Tangent, Chord, and Arc



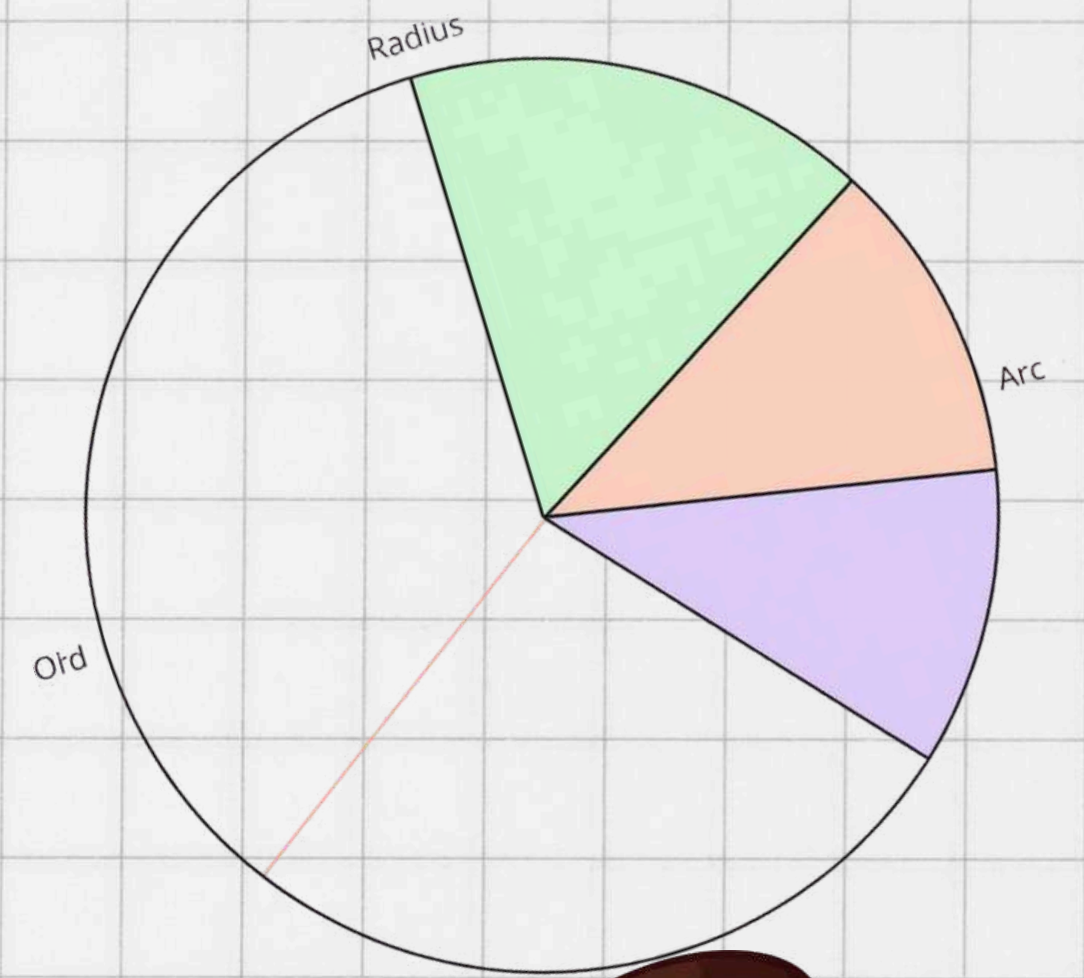
What is a Circle?



A circle is a round shape where every point on its edge is the same distance from the center. The center is the middle point, the radius is the distance from center to edge, and the diameter is the distance across the circle through the center.



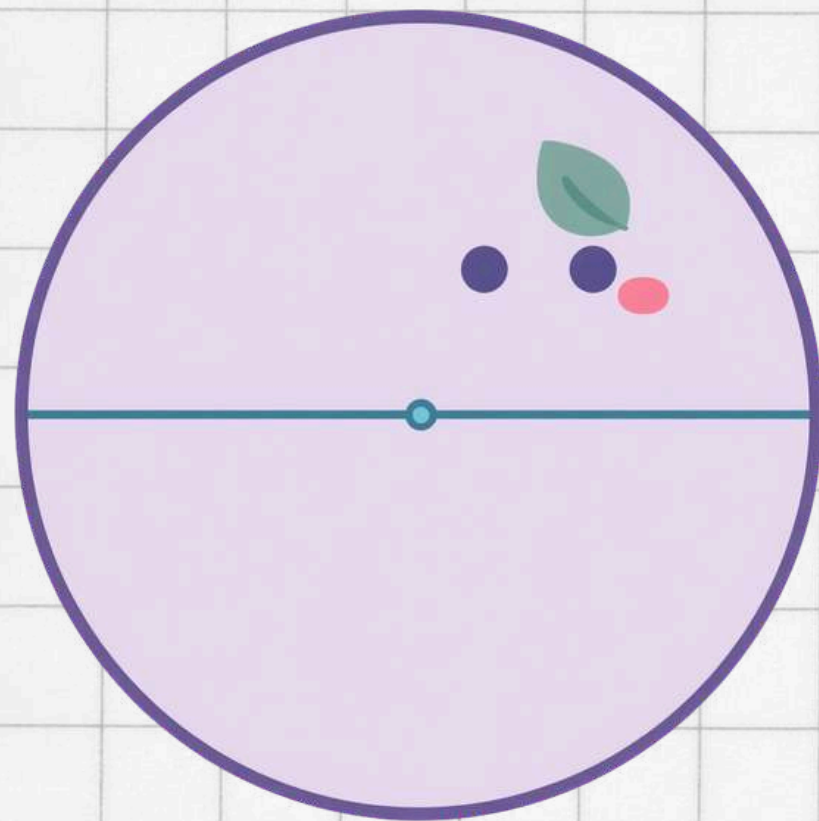
Key Parts of a Circle



Every circle has important parts to know: the radius (center to edge), diameter (across through center), chord (line joining two points), arc (curved part of the edge), and tangent (line touching at one point).



What is a Tangent?

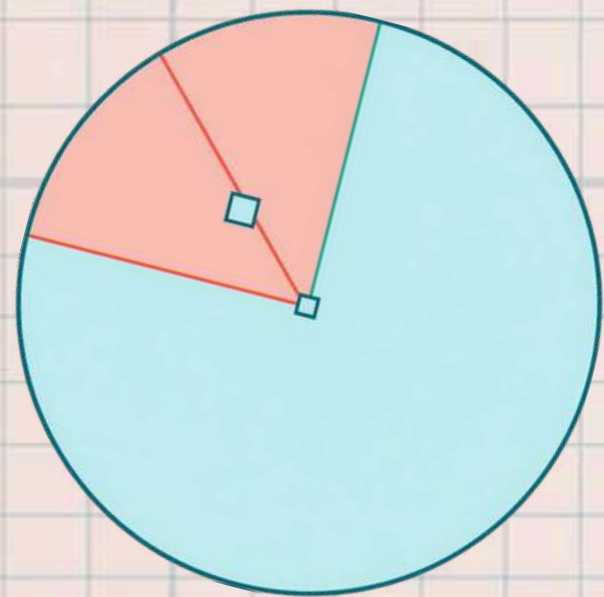


A tangent is a special line that touches a circle at exactly one point. This point where the tangent meets the circle is called the point of contact. The tangent never crosses into the circle—it just gently touches it!

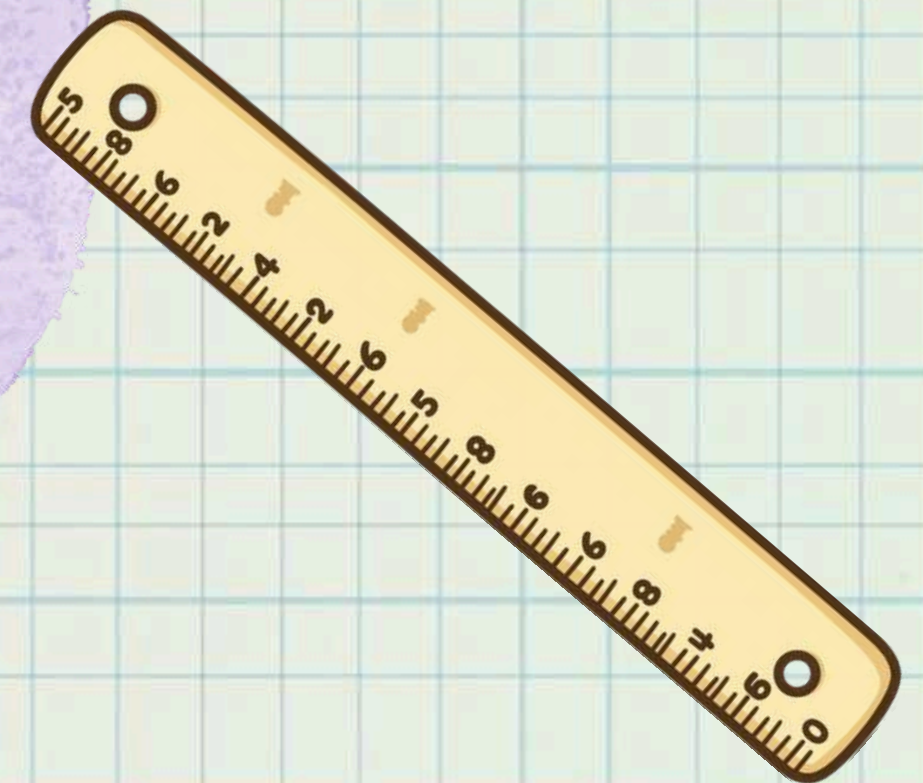


Tangent Theorem: Perpendicular to Radius

900

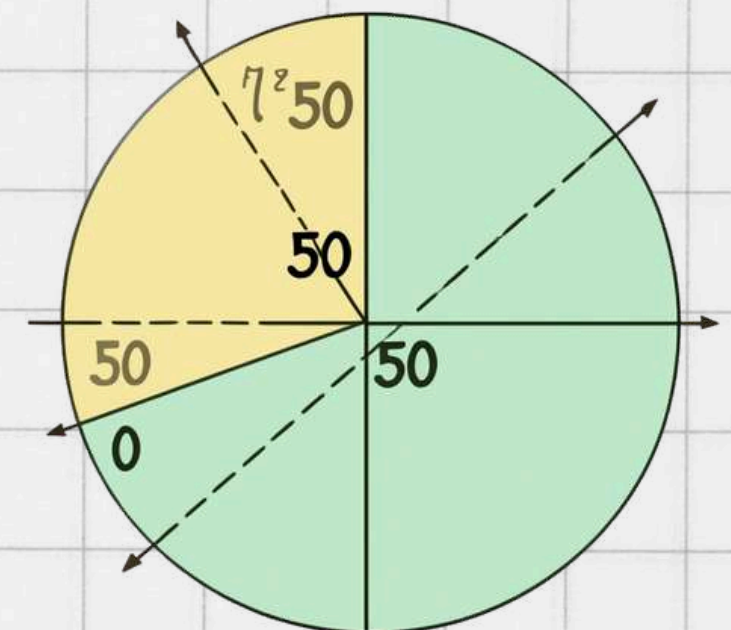
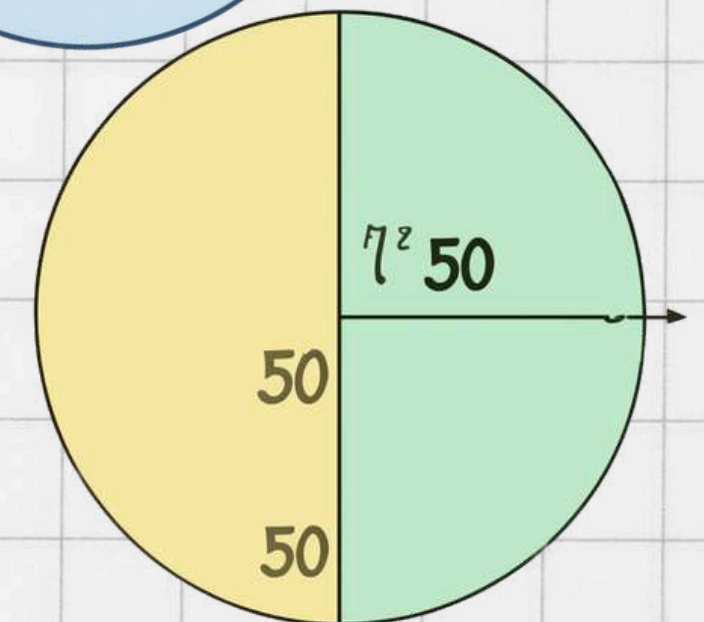
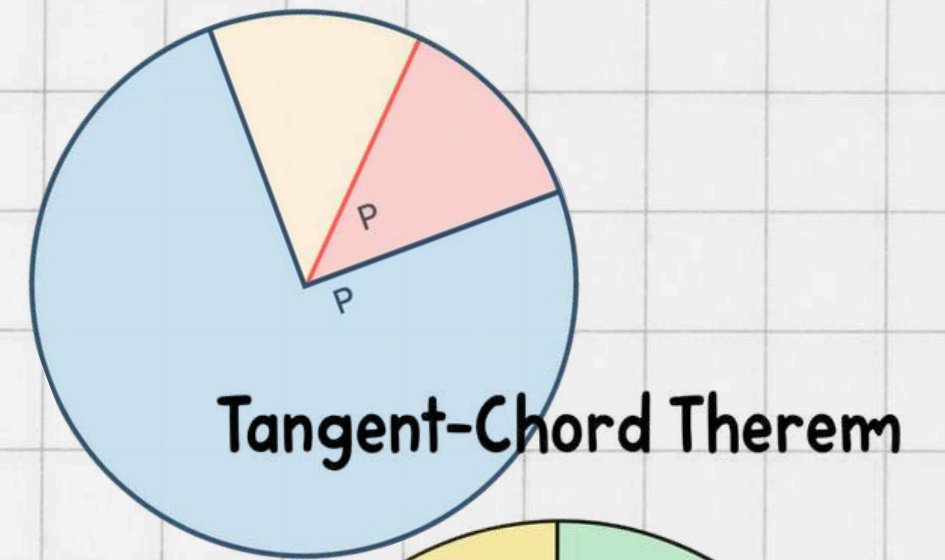


The tangent to a circle is always perpendicular to the radius drawn to the point of contact. This means they meet at exactly 90 degrees!

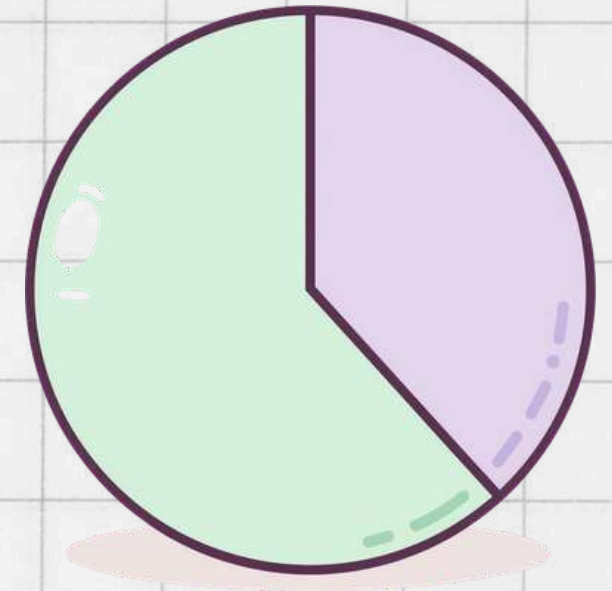


Tangent-Chord Angle Theorem

The angle between a tangent and a chord drawn from the point of contact equals the angle in the alternate segment. This means if you draw a tangent at point P and a chord PQ, the angle between them equals the angle at any point on the opposite arc!



Practice: Tangent Theorems

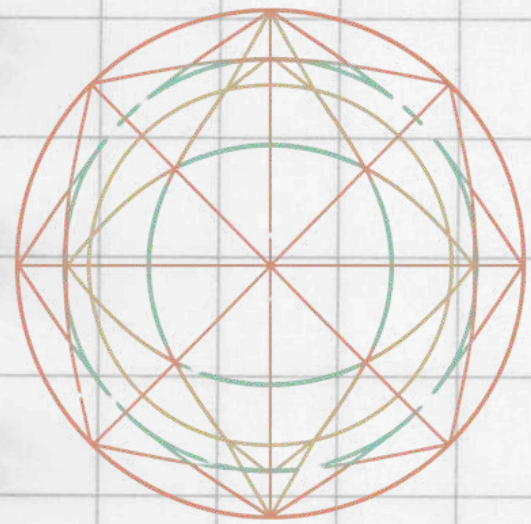


Q1: A tangent touches a circle at point P. If the radius to P is 5 cm, what angle does the tangent make with the radius?

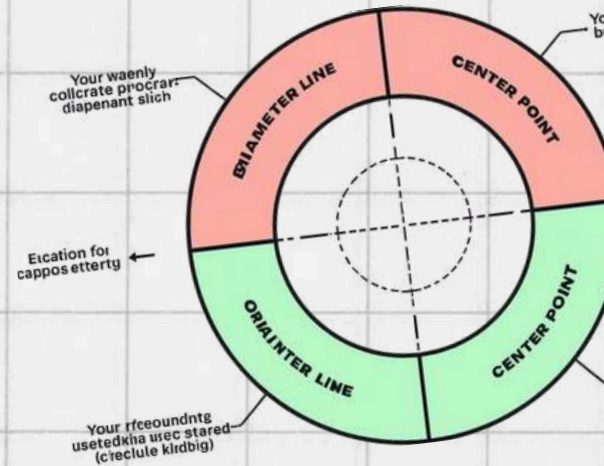
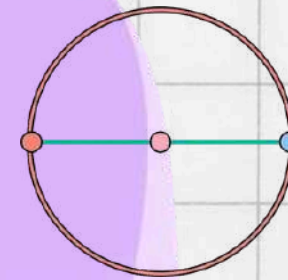
Q2: Find the angle x in the alternate segment if the tangent-chord angle is 35° .

Q3: A tangent and chord meet at point A. If the inscribed angle is 50° , what is the tangent-chord angle?

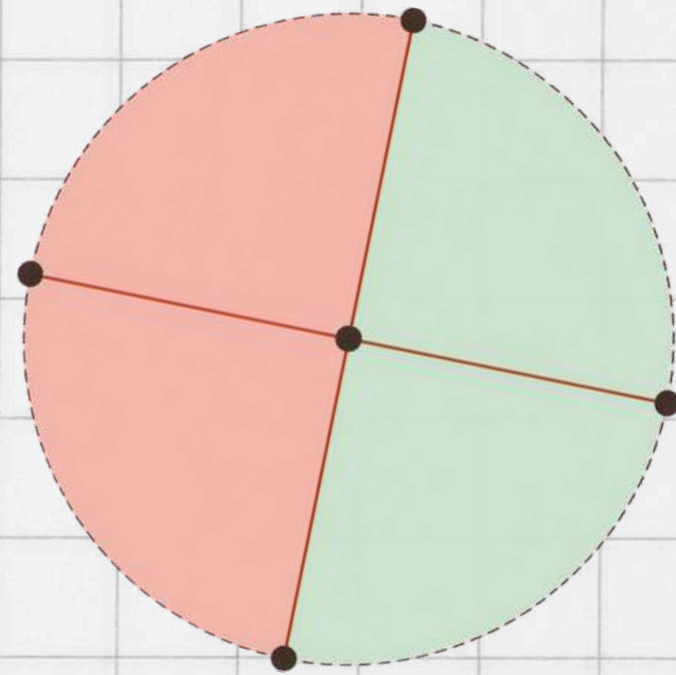
What is a Chord?



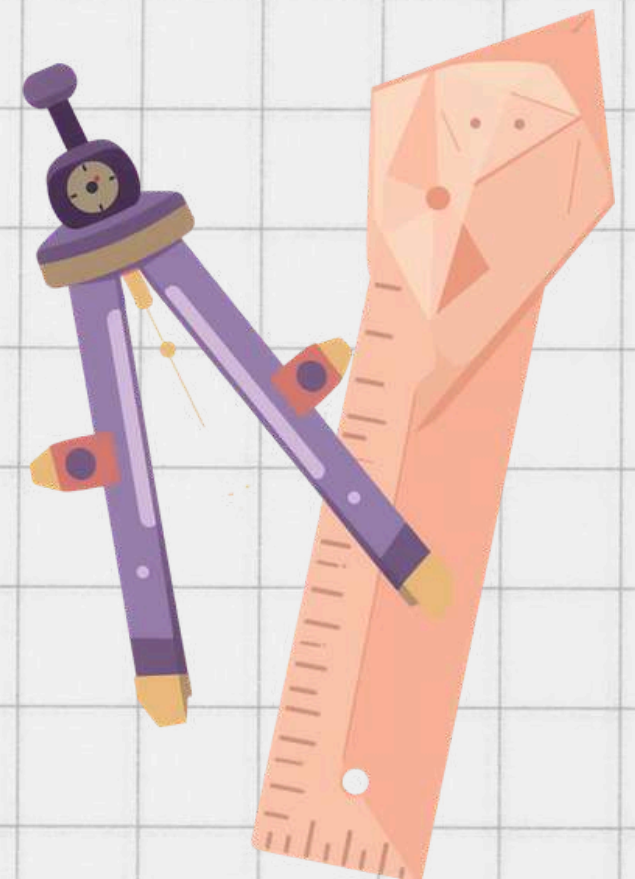
A chord is a line segment that joins any two points on a circle. The diameter is a special chord that passes through the center of the circle, making it the longest possible chord.



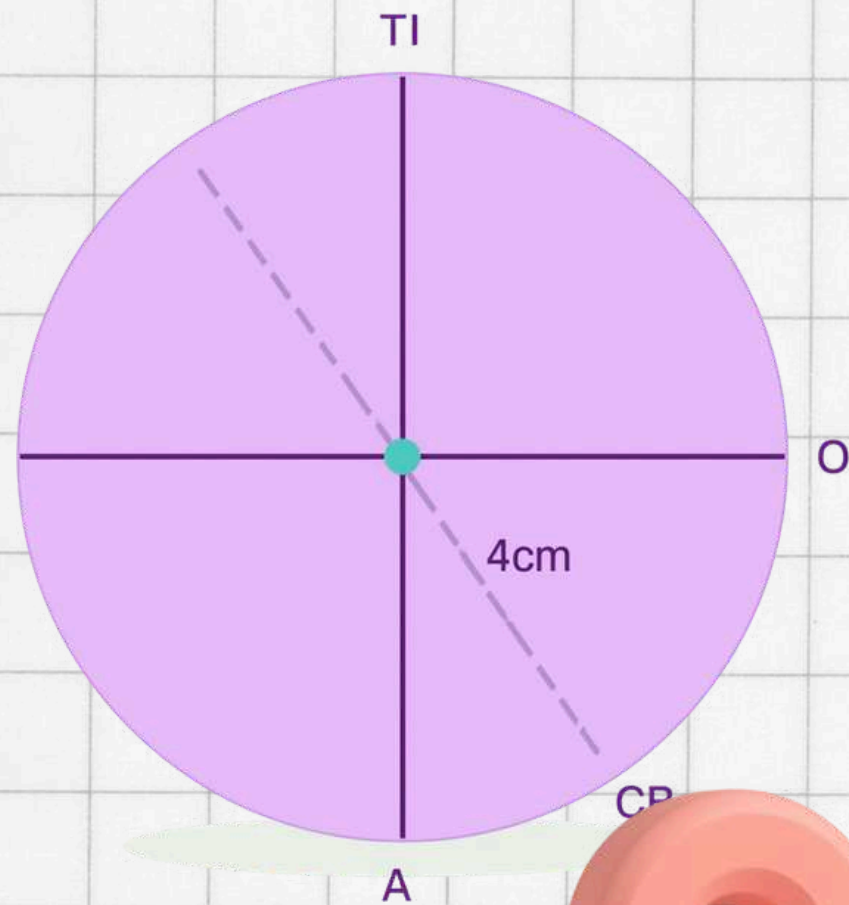
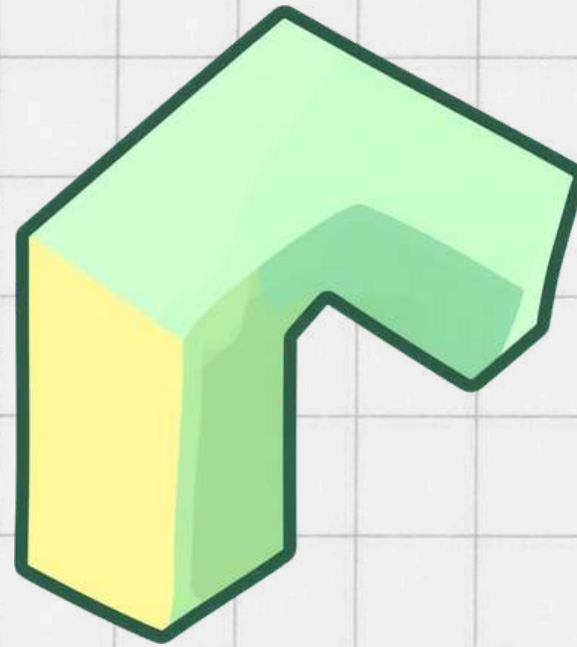
Properties of Chords



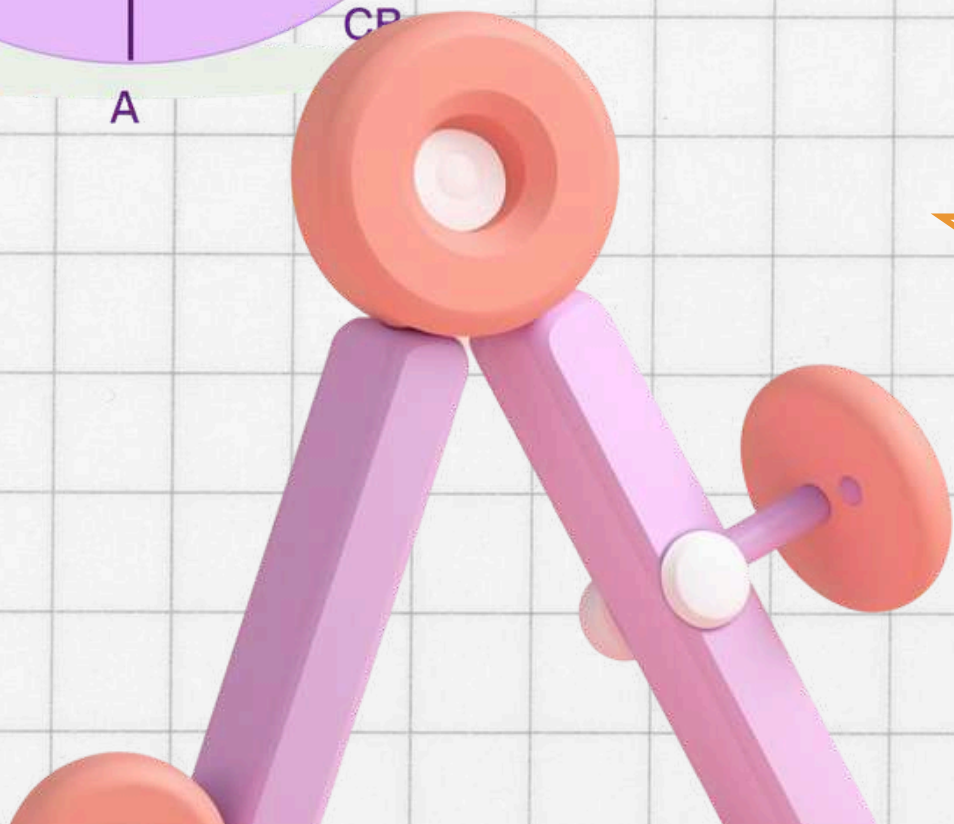
Equal chords in a circle have equal arcs! When two chords are the same length, they cut off arcs of equal size. Also, equal chords are always equidistant from the center of the circle.



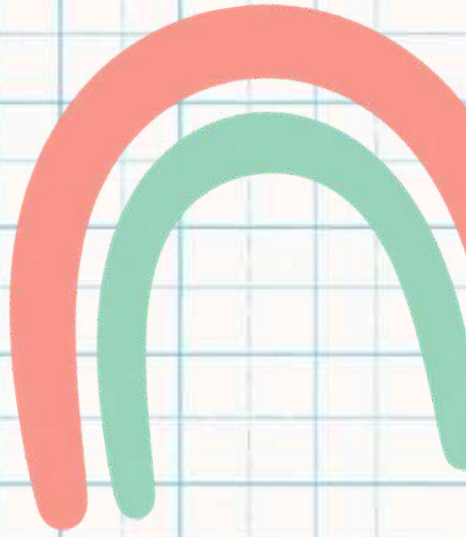
Perpendicular Bisector of a Chord



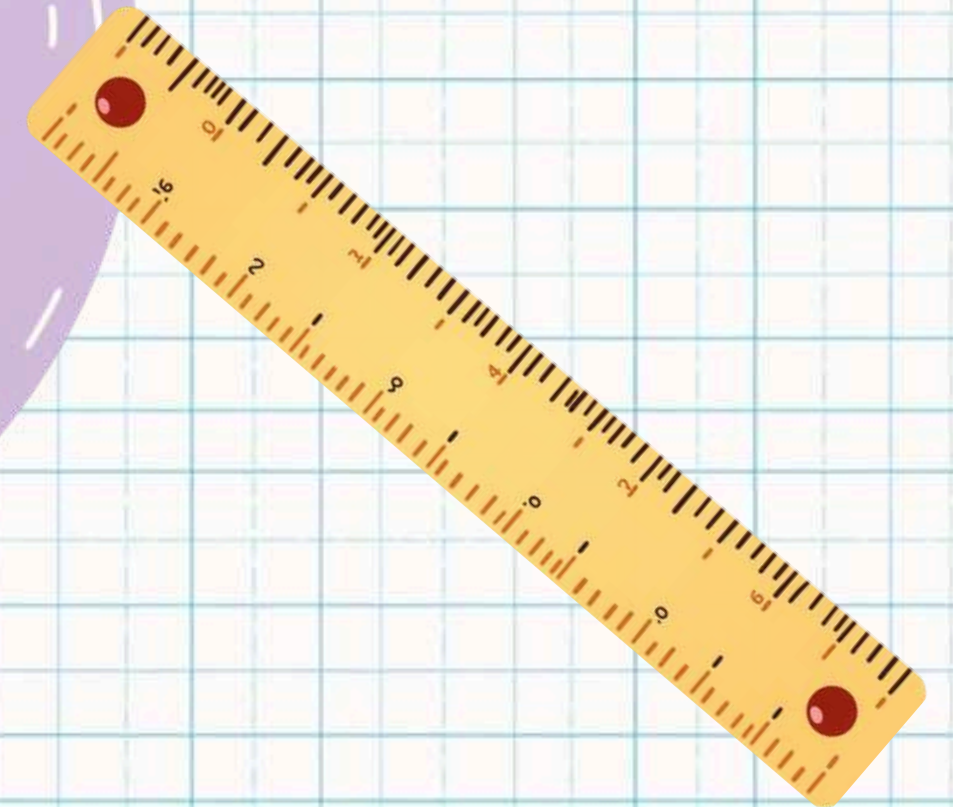
The perpendicular from the center of a circle to a chord bisects the chord into two equal parts. If $OC \perp AB$, then $AC = CB$. This theorem helps us find chord lengths and distances from the center.



What is an Arc?

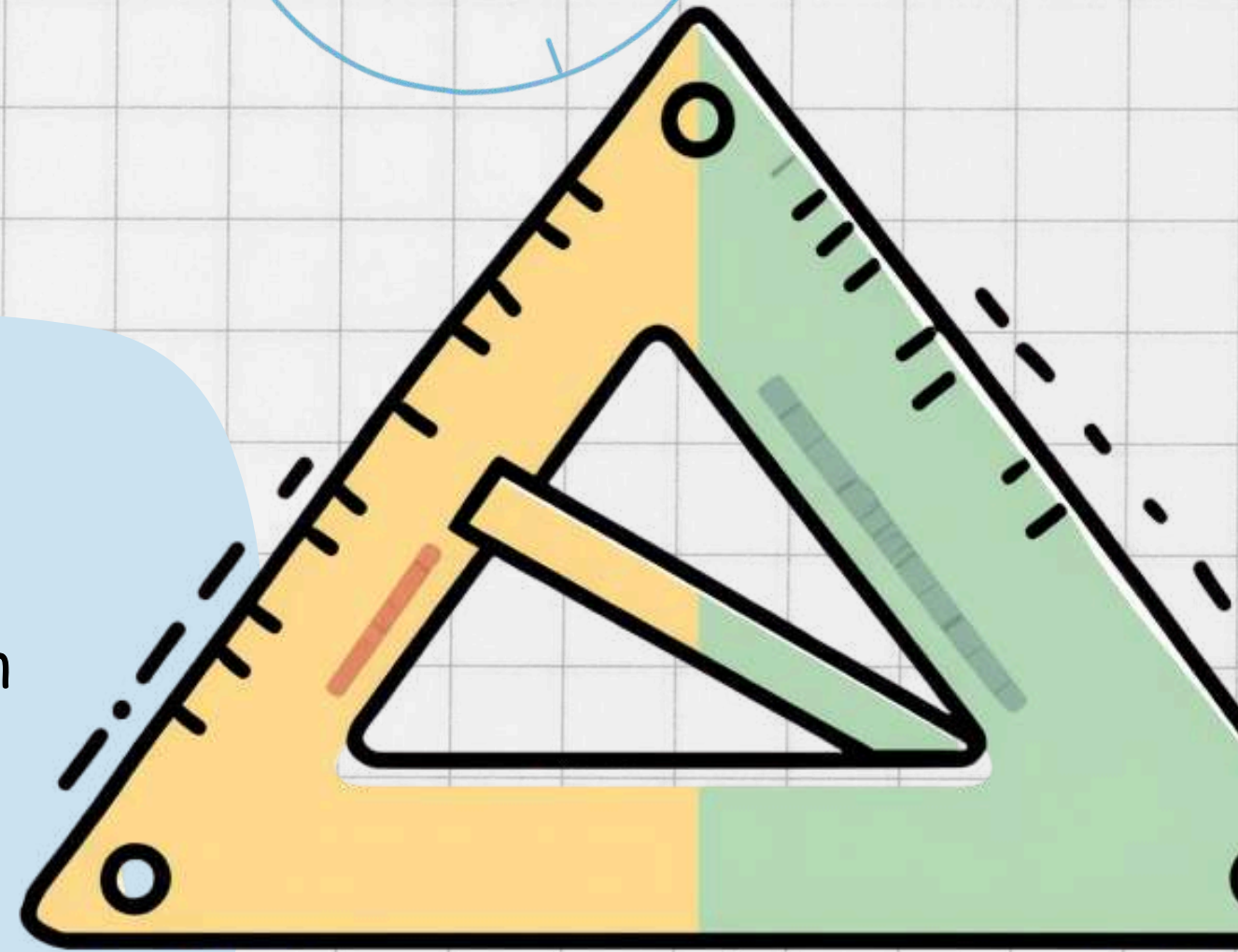
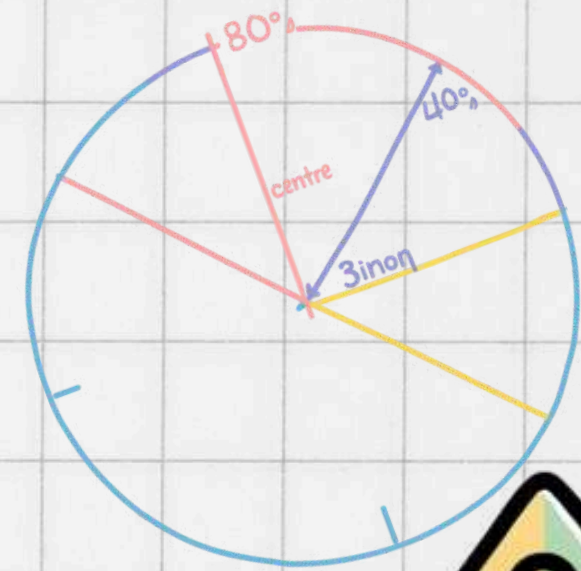


An arc is a part of the circumference of a circle between two points. A minor arc is the shorter arc, while a major arc is the longer arc. Together, they make the complete circle boundary!



Angle at the Centre and Circumference

The angle at the centre of a circle is always twice the angle at the circumference when both angles are subtended by the same arc. For example, if the angle at the circumference is 40° , the angle at the centre will be 80° !



Practice: Arc Theorems



Problem 1: If the angle at the center is 80° , what is the angle at the circumference on the same arc?

Problem 2: An angle at the circumference is 35° . Find the angle at the center subtended by the same arc.

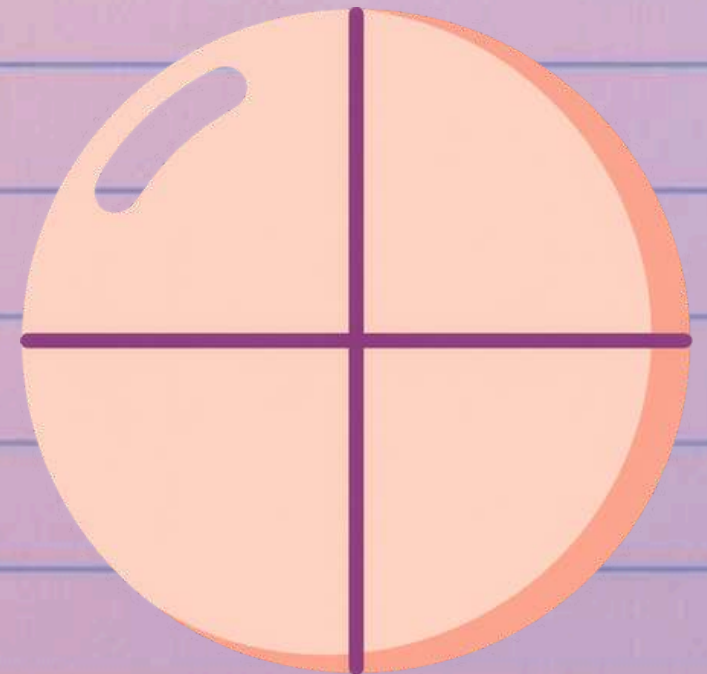
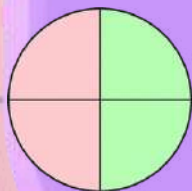
Problem 3: Two angles at the circumference subtend the same arc. One is 45° . What is the other?



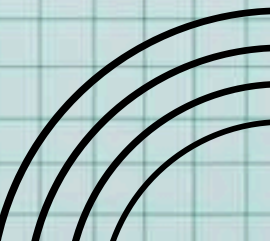
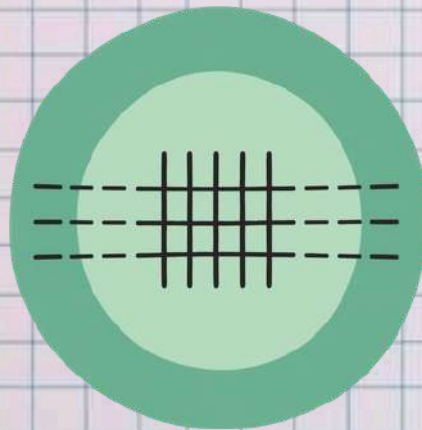
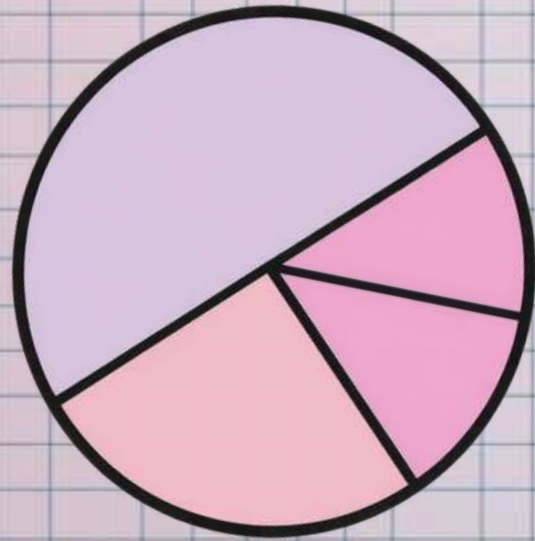
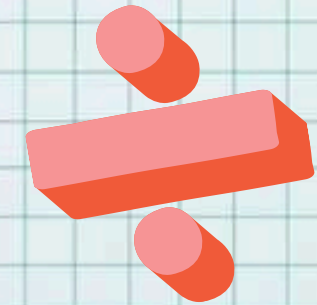
Summary of Circle Theorems



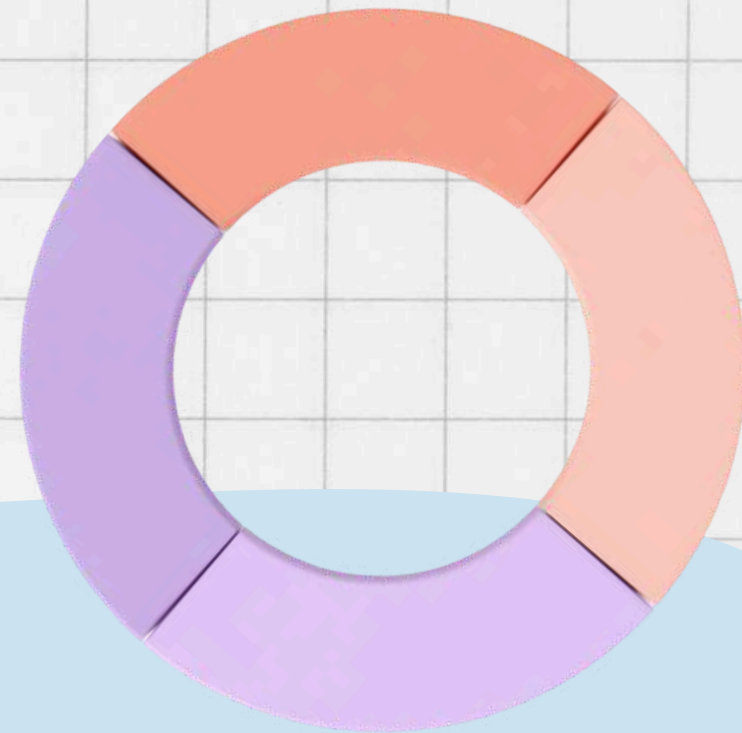
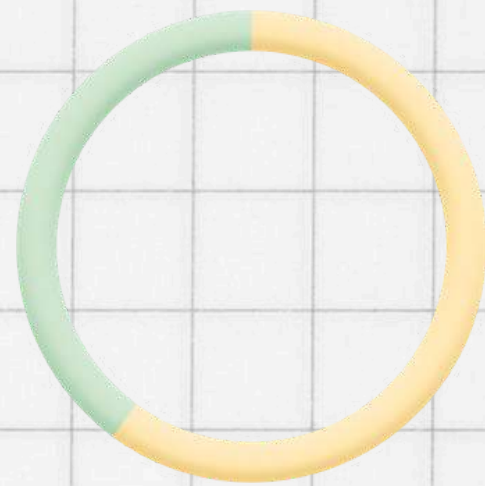
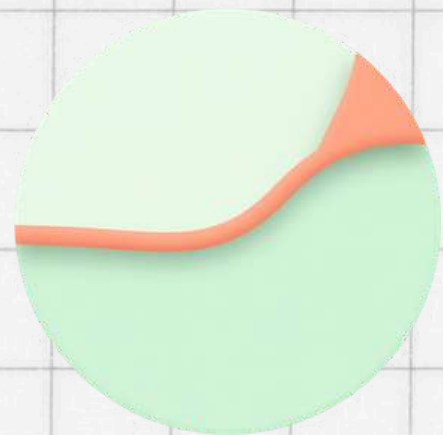
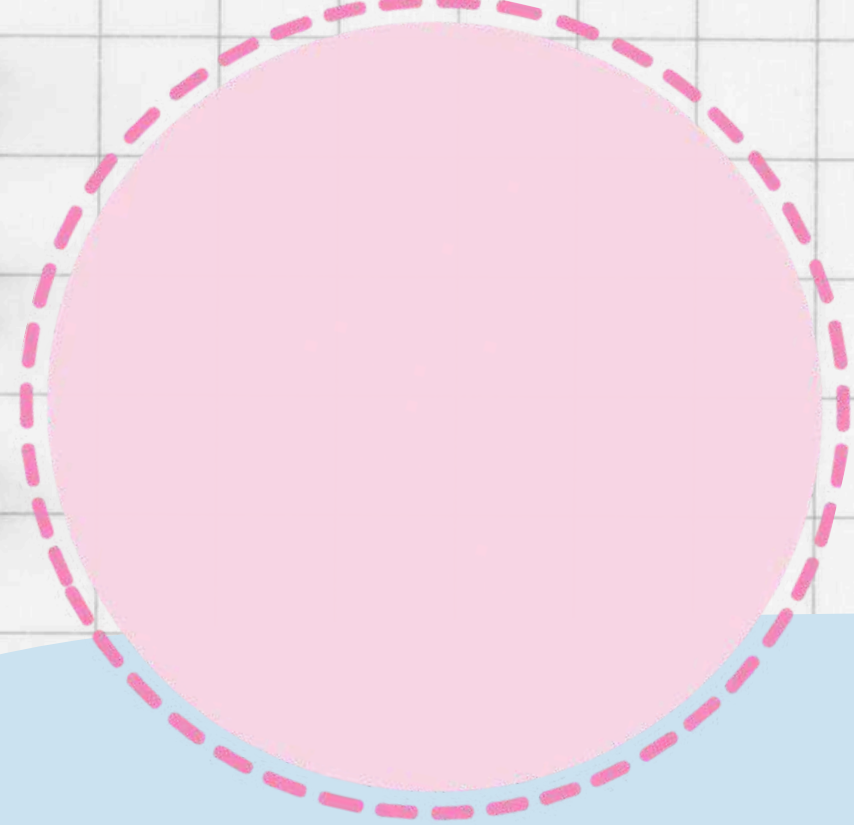
- Tangent touches circle at one point and is perpendicular to radius
 - Angle between tangent and chord equals angle in alternate segment
- Equal chords have equal arcs and are equidistant from center
 - Perpendicular from center bisects the chord
 - Angle at center is twice the angle at circumference



Challenge Problems



1. A tangent PQ touches a circle at point T . If the angle between chord TA and tangent PQ is 55° , find the angle in the alternate segment.
2. Two equal chords AB and CD are 8 cm from the center. If the radius is 10 cm, find the length of each chord.
3. An arc subtends 120° at the center. What angle does it subtend at any point on the remaining circumference?



**Thank You & Keep
Exploring Circles!**
Questions? Let's discuss!

